



CLUSTER PLACEMENT GUIDE

Guidance on Spacing Flower Turbines

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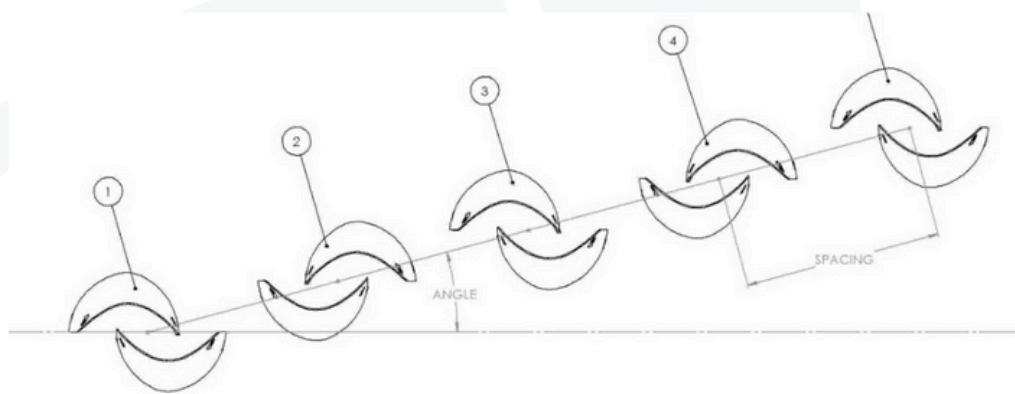
There are 3 components to placing the turbines for the cluster effect.

1. **Rotation.** They should rotate alternately clockwise and counter-clockwise. Even if adjacent ones rotate in the same direction, they will still do very well.
2. **Spacing.** For spacing, one starts with the diameter in the horizontal plane found in the spec sheet.

The ideal spacing is found by multiplying the diameter by 1.25. That is the distance between the center of one shaft to the center of the next shaft. It also works well at a multiplication factor of 1.1 to 1.3. One should use the ideal configuration but if there is a space limit, one has some freedom to vary the distances and angles and still get a good result.

3. **Angle** to the prevailing wind: The ideal is 15 degrees. This also works well from 0 to 30 degrees.

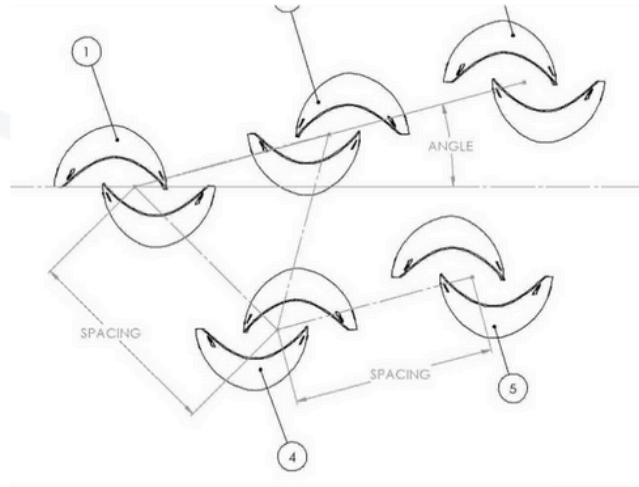
Here is how a group of 5 small ones would look when the wind comes from directly above or below:



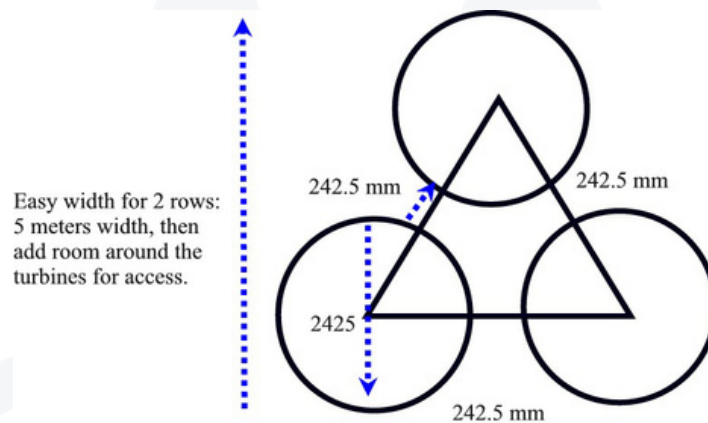
A single long row is the best choice. The same distances and angles would work in an arrangement of two rows. Two rows near each other are not recommended unless the wind is at least 5 meters per second. This is an alternative that also works well:



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The following picture shows how to arrange the turbines in a project for two adjacent rows:



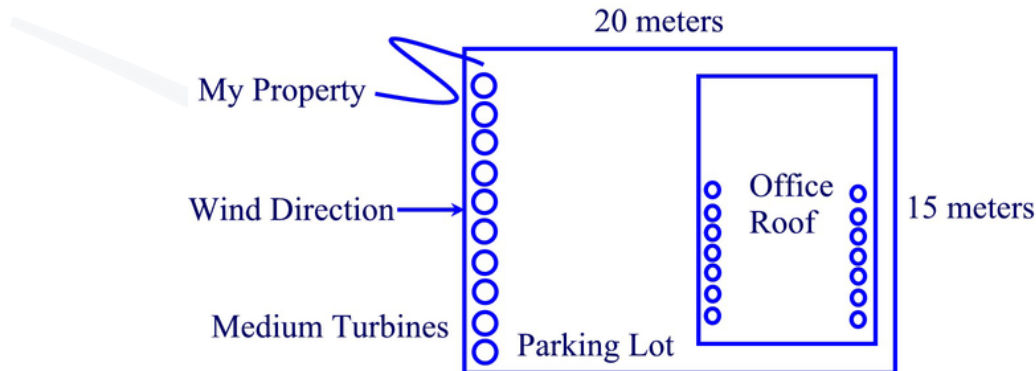
In making a project on the ground, one should allow space for an outer fence. Chain links allow wind to pass through.

The following picture shows that a good choice is making two separate rows. Ideally they should be separated by 5 times the turbine diameter, with 3 times the turbine diameter as a minimum. The reason is that the wind needs space to recover its speed after being partially consumed by the first line of turbines.

Note that the space in between the two rooftop rows, arranged in a north-south direction, would be a good location for south-facing solar panels.



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Here is a guide of how to implement the choices in this picture step by step.

Step 1: Get a bird's-eye view of the roof with distances and showing obstacles like an HVAC system or elevator shaft.

Step 2: Obtain the wind speed and direction and place an arrow with the predominant direction on the picture of the roof. This is important so we can maximize the bouquet effect, which is ideally around 15-20 degrees either way off the perpendicular line to the wind direction, but it works up to 45 degrees off the perpendicular line in either direction.

Step 3: Pick the size of turbine that is safe to place on the roof. The larger the better. Most projects will pick our 2-meter blade height turbine.

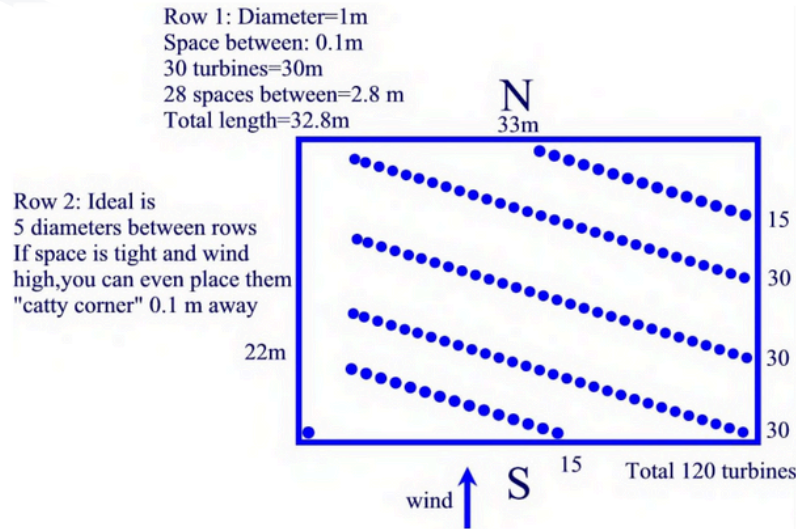
Step 4: Our turbines are usually built with a diameter half of the height. That diameter times 0.1 is the closest you can put the turbines. So a row of 10 turbines will require approximately 10 meters for the turbines and 9 spaces in between, or 0.9 meters. Choose a way to fit them in rows with the ideal direction.

Step 5: You can have multiple rows, but it is best to separate the rows by three to five times the blade diameter. Using the 2-meter turbine with a diameter of 1 meter, one would separate the rows by 3 to 5 meters. The reason for this is that, even though a row placed close behind another experiences a clustering effect, some of the energy is used up in the first row and less is available for the second row. Spacing the turbines in a row apart from the next row allows the wind to recover its strength.

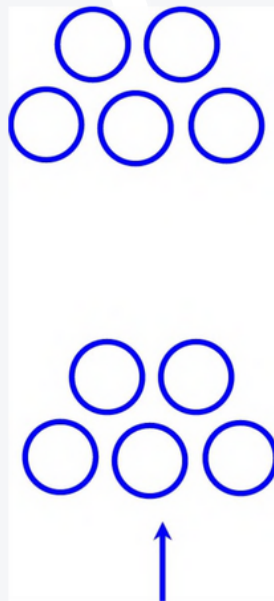


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Here is a different example of how to make multiple rows on a roof:



One can also make two adjacent rows followed by more two adjacent rows with a separation of 5 times the diameter of the turbine. This is usually recommended only for wind of speed at least 5.5 meters per second. The reason for this higher wind threshold than earlier is that the two rows together absorb more of the wind's energy and the recovery to the wind speeds that originally impacted the turbines is harder.





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Although our work is original and patented, you may find it helpful to think of this advance in wind power as a similar concept with birds and airplanes flying together:

